

Wenbo Lyu

B.Sc. Graduate

Curriculum Vitae

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Education and qualifications

B.Sc. In Geographic Information Science Shaanxi Normal University 2025

Research Experience

2026.6- Research Assistant, The Hong Kong Polytechnic University
2025.9-2026.6 Research Assistant, Eastern Institute of Technology, Ningbo
2024.8-2025.8 Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

Research interests

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.

Selected research papers

1. Lyu, W, S Dai, Y Song, W Zhao, W Yi, Y Xiao, and N Jia (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*, 1–23.
2. Lyu, W, Y Lei, W Yi, Y Song, X Li, S Dai, Y Qin, and W Zhao (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems* **127**, 102435.
3. Lv, W, Y Lei, F Liu, J Yan, Y Song, and W Zhao (2025). gdverse: An R Package for Spatial Stratified Heterogeneity Family. *Transactions in GIS* **29**(2), e70032.
4. Lv, W, F Liu, K Cai, Y Cao, M Deng, W Liang, J Yan, and G Wang (2024). Distinguishing the Impacts and Gradient Effects of Climate Change and Human Activities on Vegetation Cover in the Weihe River Basin, China. *Journal of Geophysical Research: Biogeosciences* **129**(10), e2024JG008297.
5. Li, B, W Lv, H Ding, Y Song, and W Zhao (2026). New-type urbanization policy, urban expansion, and employment quality in chinese cities: evidence from a quasi-natural experiment in China. *GIScience & Remote Sensing* **63**(1), 2716413.
6. Xu, L, H Shi, W Lyu, Y Zhang, and H Zhao (2026). Beyond correlation: Unveiling the causal link between street-scapes and recreational walking/cycling. *Journal of Transport Geography* **136**, 104789.
7. Chen, C, Y Song, W Lv, A Shemery, K Hampson, W Yi, Y Zhong, and P Wu (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering*.

Note: Publications under “Lyu, W.” and “Lv, W.” both refer to Wenbo Lyu.